

Nutritional Source Cues:

Social Media Nutrition Credibility

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Table of contents

1	Introduction	2
1.1	Research Question	2
1.2	Hypotheses	2
2	Background/Literature Review	3
3	Methodology	4
3.1	Study Design	4
3.2	Outcome Measures and Covariates	5
3.3	Sample and Procedure	5
3.4	Randomization and Design Checks	6
4	Results	7
4.1	Observational Trends	7
4.2	Models	8
4.3	Regression Results	9
4.4	Diagnostic Checks	14
5	Discussion	18
6	Conclusion	19
7	References	20
8	Appendix	21
9	Appendix A: Survey Flow	21
	Appendix B: Treatment	22
	Appendix C: Survey	26

1 Introduction

Social media has become a primary avenue for many people to consume information such as news, health advice, entertainment, and many other topics. With the use of AI, social media has become more influential in targeting individuals with advertising and other information. To the most extent, information related to entertainment, advertisements might be okay. However, when individuals are exposed to health-related information, not only experts in this field are sharing their perspective, we are also seeing many influencers sharing their perspective based on their experience, their observations, etc.

While often influencers generate higher volume of engagement and visibility, experts are mostly considered as credible due to their knowledge, education, and expertise. This raises an important question, does the source of information itself shape how the information is perceived by users and how do they act on this advice?

In real-world contexts, it's challenging to determine the effect of the source because posts from experts and influencers vary across multiple dimensions, including varying audiences, and presentation styles. Consequently, observational comparisons cannot accurately assess whether differences in engagement or credibility are genuinely influenced by the source or if they are due to these confounding factors.

To address this challenge, this experiment utilizes a randomized controlled experiment that keeps the content of a nutritional message consistent while varying only the source label (either expert or influencer). By randomly assigning participants to different conditions, the design isolates the causal impact of source cues on key outcomes, such as perceived credibility and trust. This method allows for a clear examination of whether labeling information as originating from an expert rather than an influencer significantly changes how individuals evaluate and respond to nutritional content.

1.1 Research Question

Are expert recommendations perceived as more credible than non-expert recommendations in the context of social media nutritional posts or recommendations?

1.2 Hypotheses

1.2.1 Null Hypothesis (H0):

There is no difference in credibility measures, such as accuracy, authenticity, and believability, between participants exposed to expert and influencer source cues.

$$H_0 : E[Y|Expert] = E[Y|Influencer]$$

1.2.2 Alternative Hypothesis (H1):

Participants exposed to the expert source cue will report higher credibility measures, such as accuracy, authenticity, and believability, compared to those exposed to the influencer source cue.

$$H_a : E[Y|Expert] > E[Y|Influencer]$$

where Y represents outcome such as perceived credibility, trust, or behavioral intention.

2 Background/Literature Review

Source credibility theory provides a useful framework for understanding why different messages could change people’s evaluation of information on social media, particularly health or nutrition recommendations. This theory suggests that individuals evaluate information based on three key factors: expertise, trustworthiness, and likeability. Pornpitakpan (2004) found that source credibility is crucial to how participants evaluate information, and expert sources are found to be more credible than non-expert sources, though these effects vary by context. A recent study upholds these findings. In an experiment using Instagram posts focused on e-cigarette education, Lee et al. (2024) found that posts attributed to health experts produced higher perceived credibility and greater trust than identical posts attributed to a friend or social media influencer.

However, current research also points to additional meaningful credibility cues not captured by source credibility theory, including both authenticity and distrust of expertise. Zhang and Lu (2023) argue that online audiences also respond to authenticity displays and parasocial interaction. In their study of scientists and influencers communicating on social media, they find that source cues affected perceptions of both authenticity and expertise, the former defined by whether a source appears genuine, self-revealing, and socially relatable. With respect to trust in expertise, Merkley (2020) found a link between anti-intellectualism and distrust of expert sources. Similarly, Case et al. (2022) found that political beliefs can shape how one evaluates expertise and trustworthiness. They tested the effect of source cues and message frames on COVID recommendations and found no effect as political ideologies explained nearly all variation in responses. These studies suggest that source credibility may derive from personal experience/authenticity or political in-group status, in addition to expertise and factual evidence.

Taken together, the literature suggests competing possibilities for how people might evaluate information on social media, such as nutritional recommendations. Expert sources

may enhance perceived accuracy and believability because they signal knowledge and trustworthiness (Pornpitakpan 2004), while influencer sources may project authenticity because they fit the relational norms of social media (Zhang and Lu 2023). This study tests these competing credibility signals using nutritional recommendations recently in the news.

3 Methodology

3.1 Study Design

This study employs a within-subject experimental design where individual subjects are exposed to four randomized Instagram posts consisting of two treatment posts with content attributed to either an expert or influencer. Treatments consist of the bio of an expert or influencer and their comments on a nutritional subject. The aim of the experiment is to estimate the average within-individual difference in perceived credibility between expert and influencer on nutritional content. The topics include two treatment conditions covering nutritional subjects (red meat and seed oils) and two filler posts (hydration and sleep).

A within-subject design was preferred in response to our power analysis results, where a between-subject design was used. Results indicated the study would be heavily underpowered due to the limited number of observations obtainable in this study. The power analysis conducted using a between-subject design with an effect size of 0.3 (based on prior studies of similar interest) required more than 1300 observations to be sufficiently powered. To improve power, a within-subject design was implemented where subjects viewed all content.

While power is improved in a within-subject design due to increased observations and leveraging within-person variation, there are a few drawbacks such as demand effects, hypothesis guessing, and subjects inferring the study’s intent. To address these issues, we increased the number of exposures from two to four by including filler content. To minimize study inference, all exposures were randomized on the condition that subjects are never exposed to both treatments consecutively.

Table 1: **Within-Subject Design Combinations**

	Red Meat	Seed Oils
Expert	Expert cue + red meat claim	Expert cue + seed oils claim
Influencer	Influencer cue + red meat claim	Influencer cue + seed oils claim

3.2 Outcome Measures and Covariates

The primary outcomes are participants' evaluations of each post's credibility, measured in perceived accuracy, authenticity, and believability. Responses were recorded on a 1–7 Likert scale, with higher values indicating greater perceived credibility. These measures were analyzed individually and combined into a composite credibility index (meat and oil expert credibility, meat and oil influencer credibility).

Pre-treatment covariates include variables such as health literacy and confidence, use of social media for health and nutrition, and political alignment, along with indices capturing subjects' baseline attitudes and behaviors on trust in experts and health engagement. The expert trust index was constructed by averaging response items related to trust in experts and influencers. Similarly, the health engagement index was constructed by averaging responses related to health behaviors and information seeking.

In addition to primary outcomes and prior belief measures, the survey recorded completion time metrics, the sequence order of post exposure, and four attention checks. The first two attention checks were administered during pre-treatment questions, and the third and fourth were presented immediately following each social media treatment post viewed by each participant. The order of post presentation was recorded to assess whether the position of posts within the sequence affects outcome measures.

These attention checks were designed to assess respondent comprehension of survey questions and identify low-effort responses. Completion time measures were similarly used to evaluate the extent to which participants may have rushed through portions of the survey. Together, these measures allow for the detection of low-quality survey responses across treatment conditions.

3.3 Sample and Procedure

Subjects were recruited through Pure Spectrum, an online panel provider. Using Pure Spectrum's demographic census data, demographic data was automatically populated and used to provide the necessary quotas for a target sample size of 250 subjects (though some quotas were exceeded in an attempt to achieve others). Qualtrics was the survey platform used to capture responses and demographics from participants redirected by Pure Spectrum.

The survey consisted of multiple blocks generated in the Qualtrics platform. First, subjects were provided an introduction to the survey along with instructions and a consent block. Following consent, subjects completed demographic questions, pre-treatment measures assessing baseline attitudes and behaviors, and two attention checks.

Once pre-treatment measures are completed, subjects are exposed to a sequence of social media posts. Each post first begins with an image of the author’s bio including their handle, title, occupation, and number of followers – and highlighting their credentials in the case of experts – followed by a post commenting on a nutritional recommendation. The filler posts also followed this pattern (bio then post). Following each post, subjects answered a set of questions evaluating the content in terms of accuracy, authenticity, and believability. The survey then concludes with an outro statement thanking participants for completing the survey.

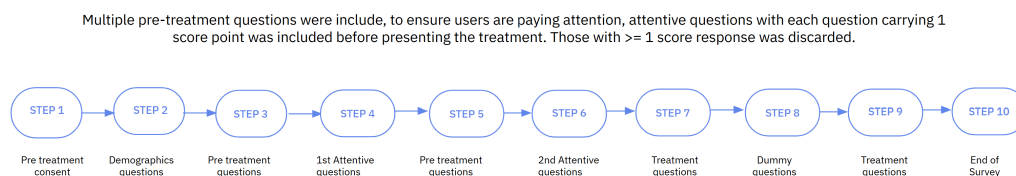


Figure 1: Survey flow diagram illustrating treatment assignment and measurement sequence

3.4 Randomization and Design Checks

The order of post presentation was randomized at the individual level. To ensure internal validity and minimize order effects, both the assignment of topic to source type and the sequence in which the posts appeared were randomized. However, to reduce hypothesis guessing treatments were also randomized in a way where they never appeared one after the other in a given sequence. As shown below, first participants were randomized with respect to whether they saw a red meat post from an expert and a seed oil post from an influencer or the reverse. Next they were randomized with respect to the order in which they saw treatment and filler posts.

Subjects were included in the analysis if they achieved a CAPTCHA score greater than or equal to 0.5, spent at least one third of the mean duration on the entire survey, and passed all attention checks. This restriction ensures that the sample consists of respondents who were not bots and engaged meaningfully with the survey content.

To assess the validity of the experimental design and evaluate potential sources of bias, we conduct a series of diagnostic checks focusing on randomization, respondent engagement, and potential order effects. First, we assess whether the randomized order of treatment exposure is independent of pre-treatment covariates. We examine whether the position of treatment within the sequence is systematically related to baseline characteristics such as

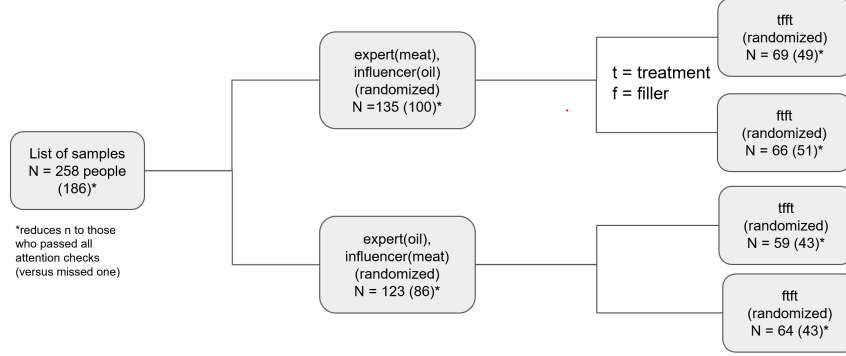


Figure 2: Randomization diagram

race, age, and education. Evidence of such relationships would indicate potential imbalances in the randomization procedure.

Second, to evaluate differential engagement across conditions, we examine whether attention and completion time vary systematically with treatment assignment and position within the sequence. These measures serve as proxies for respondent effort and allow us to detect whether certain conditions induce lower engagement or rushed responses.

Finally, we assess whether the position of posts within the sequence affects outcome measures. Because participants are exposed to multiple treatments, responses to later posts may be influenced by fatigue, learning, or prior exposure. To account for this, we examine whether outcomes vary with position and whether treatment effects differ depending on when a post appears in the sequence.

4 Results

4.1 Observational Trends

Before we dive into the modeling results, we can look at the raw data to get a baseline approximation of our results. The eye test shows us a slight bump in credibility for expert claims compared to influencer claims for both red meat and seed oil examples. The difference does look small, and our sample size is less than 200 since we limited our population to compliers only, so let's see what the regression results show.

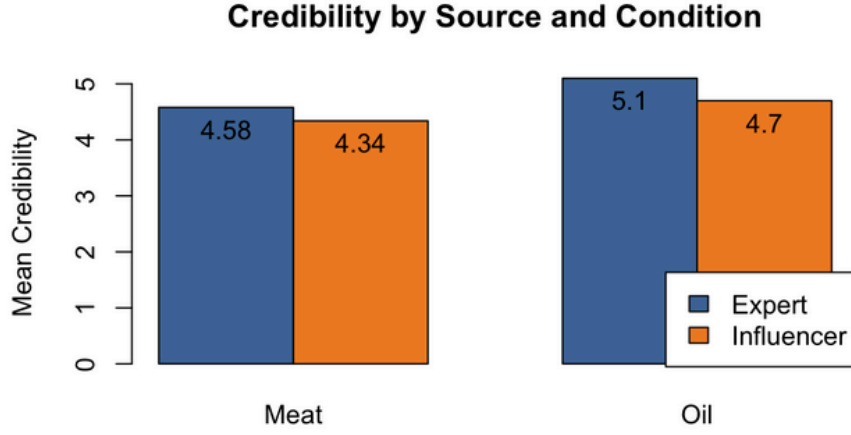


Figure 3: Credibility by Source and Condition

4.2 Models

We ran four different models to measure our causal effects, each one increasing in complexity. They included:

- A plain model with no covariates:

$$\text{Credibility}_i = \beta_0 + \beta_1 \text{Influencer}_i + \varepsilon_i \quad (1)$$

- A model with prior indexes:

$$\begin{aligned} \text{Credibility}_i = & \beta_0 + \beta_1 \text{Influencer}_i \\ & + \beta_2 \text{ExpertTrustIndex}_i \\ & + \beta_3 \text{HealthEngagementIndex}_i + \varepsilon_i \end{aligned} \quad (2)$$

- A model with prior indexes and demographic covariates:

$$\begin{aligned} \text{Credibility}_i = & \beta_0 + \beta_1 \text{Influencer}_i \\ & + \beta_2 \text{ExpertTrustIndex}_i + \beta_3 \text{HealthEngagementIndex}_i \\ & + \beta_4 \text{Age}_i + \beta_5 \text{Gender}_i + \beta_6 \text{Education}_i \\ & + \varepsilon_i \end{aligned} \quad (3)$$

- A model with prior indexes and an interaction term:

$$\begin{aligned}\text{Credibility}_i &= \beta_0 + \beta_1 \text{Influencer}_i \\ &\quad + \beta_2 \text{HealthEngagementIndex}_i \\ &\quad + \beta_3 (\text{Influencer}_i \times \text{HealthEngagementIndex}_i) \\ &\quad + \varepsilon_i\end{aligned}\tag{4}$$

4.3 Regression Results

We've included three analyses and results tables below:

4.3.1 Red Meat Claims

Table 1: Meat Regressions

	<i>Dependent variable:</i>			
	Meat Credibility			
	(1)	(2)	(3)	(4)
Influencer	-0.239 (0.244)	-0.368 (0.237)	-0.371 (0.235)	-2.331* (1.311)
Expert Trust Index		0.192 (0.157)	0.191 (0.160)	
Health Engagement Index		0.396*** (0.127)	0.425*** (0.127)	0.332** (0.143)
Age 18-24			0.154 (0.447)	
Age 25-34			-0.764 (0.469)	
Age 35-44			0.249 (0.468)	
Age 45-54			0.100 (0.442)	
Age 55-64			0.482 (0.436)	
Gender			-0.021 (0.240)	
Education			-0.085 (0.108)	
Influencer*Health Engagement				0.363 (0.239)
Constant	4.580*** (0.166)	1.565* (0.794)	1.619* (0.938)	2.852*** (0.764)
Observations	186	186	186	186
R ²	0.005	0.093	0.148	0.097
Adjusted R ²	-0.0002	0.078	0.099	0.082
Residual Std. Error	1.660 (df = 184)	1.594 (df = 182)	1.575 (df = 175)	1.590 (df = 182)
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

Figure 4: Red Meat Regression

Starting with the causal effect on red meat claims, we do see an effect between influencers and experts, but the results are not statistically significant. As we added covariates, our R^2 goes up, which is a good sign, since the model is doing a better job of capturing

the variation in the data, but the treatment effect remains statistically insignificant. Our standard errors do go down though, meaning our actual treatment effect region narrows, indicating that there might be an effect there, but that our dataset is simply underpowered. We do see a statistically significant effect for being previously engaged in health content via the Health Engagement Index though, showing that prior biases do affect credibility.

4.3.2 Seed Oil Claims

Table 2: Oil Regressions

	<i>Dependent variable:</i>			
	Oil Credibility			
	(1)	(2)	(3)	(4)
Influencer	-0.397** (0.201)	-0.322 (0.198)	-0.327 (0.198)	1.456 (1.097)
Expert Trust Index		0.220* (0.131)	0.147 (0.135)	
Health Engagement Index		0.203* (0.106)	0.201* (0.107)	0.490*** (0.160)
Age 18-24			0.991*** (0.376)	
Age 25-34			0.404 (0.395)	
Age 35-44			0.700* (0.394)	
Age 45-54			0.403 (0.371)	
Age 55-64			0.502 (0.367)	
Gender			0.072 (0.202)	
Education			-0.005 (0.091)	
Influencer*Health Engagement				-0.328 (0.200)
Constant	5.097*** (0.147)	2.871*** (0.684)	2.617*** (0.813)	2.404*** (0.892)
Observations	186	186	186	186
R ²	0.021	0.078	0.121	0.077
Adjusted R ²	0.016	0.063	0.071	0.062
Residual Std. Error	1.364 (df = 184)	1.330 (df = 182)	1.325 (df = 175)	1.331 (df = 182)
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

Figure 5: Seed Oil Regression

Moving to the seed oil claims, much of the results from the red meat claims are mirrored here. The basic model does seem to have a statistically significant treatment effect on first pass, but our R^2 is low, and when we add more covariates, the statistical significance of

the effect goes away. We do see a statistically significant effect on both prior bias indexes though (Expert Trust Index and Health Engagement Index), fueling our claim that these biases impact credibility.

4.3.3 Pooled Claims with Internal Fixed Effects

Pooled Within-Subject Regressions (Meat Oil)			
	<i>Dependent variable:</i>		
	Perceived Credibility		
	(1)	(2)	(3)
Influencer	-0.306* (0.171)	-0.306* (0.171)	0.161 (0.786)
Expert Trust Index		0.417*** (0.000)	
Health Engagement Index		0.250*** (0.000)	0.627*** (0.073)
Influencer Health Engagement			-0.087 (0.147)
Constant	5.070*** (0.086)	1.670*** (0.086)	1.569*** (0.393)
Respondent FE	Yes	Yes	Yes
Respondent FE	Yes	Yes	Yes
Observations	372	372	372
R ²	0.716	0.716	0.717
Adjusted R ²	0.431	0.431	0.429
Residual Std. Error	1.167 (df = 185)	1.167 (df = 185)	1.168 (df = 184)
<i>Note:</i>		* p<0.1; ** p<0.05; *** p<0.01	

Figure 6: Pooled Within-Subject Regressions

Lastly, since we were able to run a within-subject experiment by showing all participants both claims, we pooled both red meat and seed oil responses while controlling for each

individual to boost the power of our analysis and evaluate the cumulative effect. It is here where we see a statistically significant treatment effect across all models. With double the responses, we have more power and do see that expert claims influence credibility. However, the effect is quite small (0.3).

4.4 Diagnostic Checks

Before we move on from our results, we ran some diagnostic tests to evaluate any bias in our data:

4.4.1 Attrition Check

We examined completion rates of the survey between various treatment group combinations and saw no statistically significant difference in attrition. This was expected since we only included those who completed the survey to begin with.

Table 1: Attrition Analysis: Treatment Predicting Survey Completion		
	<i>Dependent variable:</i>	
	Meat Source	Completed (Binary) Oil Source
	(1)	(2)
Influencer	−0.007 (0.026)	
post_oil_sourceinfluencer		0.007 (0.026)
Constant	0.937*** (0.018)	0.931*** (0.019)
Observations	380	380
R ²	0.0002	0.0002
Adjusted R ²	−0.002	−0.002
Residual Std. Error (df = 378)	0.249	0.249
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 Robust (HC1) SEs. Dependent variable = 1 if completed survey.		

Figure 7: Attrition Analysis

4.4.2 Attention & Time Checks

Before we limited our population to only those who passed both attention checks, we did see evidence of a difference between those who completed all, some, or none of the attention checks. This solidified our decision to drop those who didn't pass both attention checks.

Table 2: Attention Check Effects

	<i>Dependent variable:</i>	
	Meat Credibility	Oil Credibility
	(1)	(2)
Failed Attention Checks (SC0)	0.180*** (0.045)	0.046 (0.042)
Constant	4.674*** (0.056)	5.010*** (0.047)
Robust SE	Yes	Yes
Observations	1,420	1,420
R ²	0.013	0.001
Adjusted R ²	0.012	0.0004
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Figure 8: Attention Check Effects

We did see a slight effect of the amount of time spent answering the survey between treatment groups, but the effect size was so small, we deemed this issue insignificant.

Table 3: Survey Duration Effects

	<i>Dependent variable:</i>	
	Meat Credibility	Oil Credibility
	(1)	(2)
Duration (seconds)	0.00000 (0.00002)	-0.00004** (0.00002)
Constant	4.513*** (0.063)	4.934*** (0.052)
Robust SE	Yes	Yes
Observations	768	768
R ²	0.00001	0.001
Adjusted R ²	-0.001	-0.0001
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Figure 9: Survey Duration Effects

4.4.3 Position Check

Positioning of the treatment in relation to the dummies should not affect credibility scores, so we also tested for this and saw no difference there either.

Table 4: Order / Position Effects

	<i>Dependent variable:</i>	
	Meat Credibility	Oil Credibility
	(1)	(2)
Order Pattern: tfft (vs ffft)	−0.107 (0.120)	0.125 (0.098)
Constant	4.568*** (0.085)	4.850*** (0.068)
Robust SE	Yes	Yes
Observations	768	768
R ²	0.001	0.002
Adjusted R ²	−0.0003	0.001
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Figure 10: Order / Position Effects

4.4.4 Randomization Order Check

Lastly, we looked to see if there was any credibility difference depending on the ordering of which type of treatment claim is shown first, and here, we did see a statistically significant difference between red meat and seed oil claims. Expert meat credibility decreases if shown the influencer claim first, while expert seed oil credibility increases if shown the influencer claim first. The doesn't seem to be an explicit reason for this, but this is something we flagged for future investigation and may imply that the subject of the claim itself may affect the credibility of the intervention.

Table 5: Randomization Balance (Condition Sequence)

	<i>Dependent variable:</i>	
	Meat Credibility	Oil Credibility
	(1)	(2)
Influencer Meat / Expert Oil	−0.270** (0.120)	0.365*** (0.099)
Constant	4.638*** (0.079)	4.746*** (0.064)
Robust SE	Yes	Yes
Observations	768	768
R ²	0.007	0.018
Adjusted R ²	0.005	0.017
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Figure 11: Randomization Effects

5 Discussion

Our findings provide limited evidence that source cues matter for credibility judgments in social media nutrition posts. In the topic-specific regressions, the differences between expert and influencer posts are generally small and not robust across models. However, when responses are pooled across both red meat and seed oil claims in our within-subjects design, influencer posts are rated as less credible ($p < .05$), suggesting a modest overall source cue effect. These findings suggest that any source effect is likely small and context-dependent. Prior beliefs also appear to play a notable role, suggesting that source cues are interpreted through respondents’ existing attitudes to expertise and health information.

Despite our findings, we still have unresolved questions. Measuring credibility is a tough task, and there are surely more covariates that affect an individual’s beliefs. We attempted to capture some of this by measuring baseline indices, but these likely tipped participants off about the focus of our survey. We also built very simple versions of these indices by just adding up response values into a cumulative number, but more thought is warranted

to better measure prior beliefs. The randomization order effect from which nutrition claim we show first is puzzling and could be a subject of a whole new experiment in its own right. As mentioned, it may indicate the topic-dependent nature of credibility judgments. With more power, we would test source cues across a greater number of topics. We also debated internally if our focus on source cues and authenticity is sufficient to capture how people make decisions about credibility given the increasing questioning of the credibility of online posts related to algorithms, politicization, and AI use.

6 Conclusion

This study examined whether expert nutritional recommendations are perceived as more credible than non-expert recommendations in the context of social media nutritional posts. Using a within-subject design, we found limited evidence that expert recommendations are perceived as having greater credibility than those of influencers. For topic-specific models, the differences between expert and influencer posts were not significant, but in the pooled within-subjects model, expert content is rated as more credible ($p < .05$). We also find that preexisting beliefs related to trust in experts/influencers and to engagement with health topics are important predictors of how respondents rate posts. This suggests that source cues operate in conjunction with prior beliefs in credibility judgments. These results suggest that while expert source cues may improve credibility, the effect is small and dependent on context. However, given the study's limited power, the study findings should be interpreted cautiously. Nevertheless, the findings have implications for public health communication: perceptions of credibility in nutrition messaging depend on both characteristics of the messenger as well as characteristics of the audience.

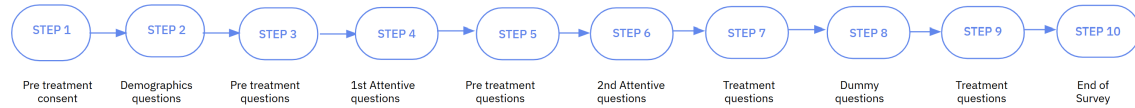
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8 Appendix

9 Appendix A: Survey Flow

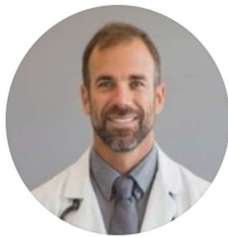
Multiple pre-treatment questions were include, to ensure users are paying attention, attentive questions with each question carrying 1 score point was included before presenting the treatment. Those with ≥ 1 score response was discarded.



Survey flow diagram illustrating treatment assignment and measurement sequence.

Appendix B: Treatment

Expert Posts



dr.jordankellogg  ...

Dr. Jordan Kellogg

1,498 posts 46.3K followers 920 following

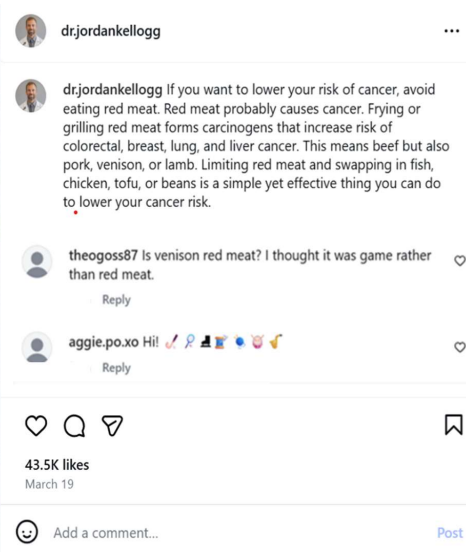
Doctor

 Harvard & Stanford-trained Gastroenterologist, San Francisco Bay Area

 Founder @snackshot.app

 4M+ Community. FREE... more

Figure 1. Expert biography, Topic: Red Meat



dr.jordankellogg

dr.jordankellogg If you want to lower your risk of cancer, avoid eating red meat. Red meat probably causes cancer. Frying or grilling red meat forms carcinogens that increase risk of colorectal, breast, lung, and liver cancer. This means beef but also pork, venison, or lamb. Limiting red meat and swapping in fish, chicken, tofu, or beans is a simple yet effective thing you can do to lower your cancer risk.

theogoss87 Is venison red meat? I thought it was game rather than red meat.

aggie.po.xo Hi!

43.5K likes
March 19

Add a comment... [Post](#)

Figure 2. Source: Expert, Topic: Red Meat



dr.priyadesai  ...

Dr. Priya Desai

1,498 posts 46.3K followers 920 following

Pediatrician

Board Certified Pediatrician. Harvard College '07, Stanford School of Medicine '14, and Northwestern Pediatrics Residency '17.

www.ohanapediatrics.com

Figure 3. Expert biography, Topic: Seed Oil



Figure 4. Source: Expert, Topic: Seed Oil

Influencer Posts



jordanliftsdaily  

Jordan Kellogg

1,939 posts 1.1M followers 1,665 following

Coach

 Health + Wellness Lifestyle

 Founder @liveelevatedlife

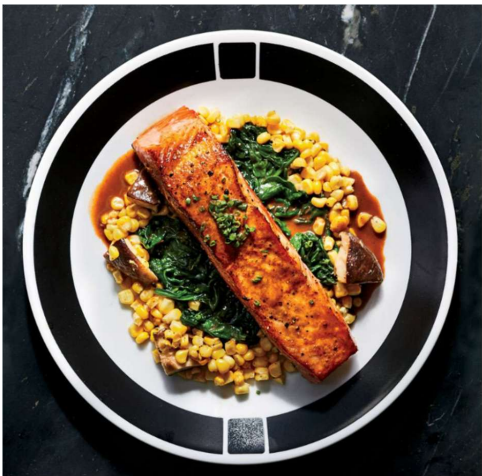
 Coach @joinladder

 Unlock Your Potential

 ... more

liveelevatedlife.com/summit-may26-apply and 2 more

Figure 1. Influencer biography, Topic: Red Meat



 jordanliftsdaily 

 jordanliftsdaily If you want to lower your risk of cancer, avoid eating red meat. Red meat probably causes cancer. Frying or grilling red meat forms carcinogens that increase risk of colorectal, breast, lung, and liver cancer. This means beef but also pork, venison, or lamb. Limiting red meat and swapping in fish, chicken, tofu, or beans is a simple yet effective thing you can do to lower your cancer risk.

 theogoss87 Is venison red meat? I thought it was game rather than red meat. 

Reply

 aggie.po.xo Hi!      

Reply

193K likes
March 20

 Add a comment... 

Figure 2. Source: Influencer, Topic: Red Meat

24



priyawellnessdaily  

PriyaWellnessDaily

1,939 posts 1.1M followers 1,665 following

Fitness Trainer

Helping women gain strength + get results that last
But also, realistic mom life 

[@bodylura](#) | [@vitaluralabs](#)

work out... [more](#)

 [join.bodylura.com](#) and 1 more

Figure 3. Influencer biography, Topic: Seed Oil

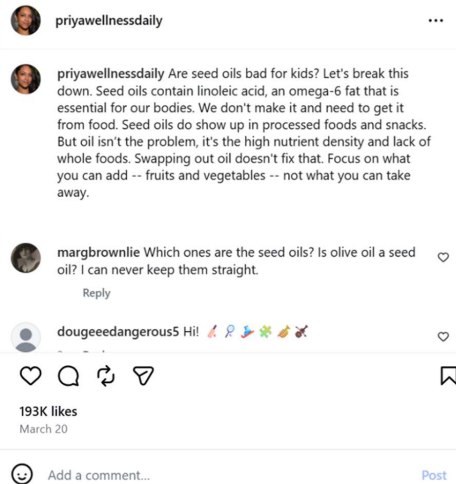


Figure 4. Source: Influencer, Topic: Seed Oil

Appendix C: Survey

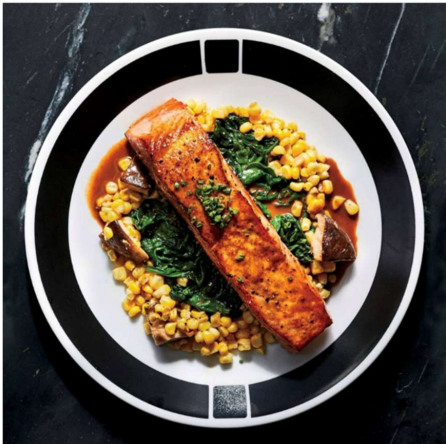
	Intro
1	Welcome! We would like your feedback as you view a short Instagram style feed. Your responses will be recorded anonymously used only for research purposes. Please click the arrow to begin.
	Consent
2	Consent to Participate What you will do You will view a short Instagram-style feed, then answer a few brief questions. Please read the posts as you normally do. Risks There is minimal risk. Some content mentions health topics. You may stop at any time by closing the survey; however, payment is provided only for completed submissions. There is no other penalty for stopping early. Privacy We collect your survey responses and your Pure Spectrum participant ID. We do not collect your name or contact information. Data will be stored securely and reported only in aggregate. Voluntary Participation Participation is voluntary. Consent By selecting “I agree,” you confirm that you are 18 years or older and consent to participate. • I agree to participate • I do not agree to participate (termination link)
2b	[if do not agree to participate] We thank you for your time spent taking this survey. Your response has been recorded.
	Demographics
3	How do you describe yourself?
	• Male • Female • Nonbinary • Prefer to self-describe • Prefer not to say
4	How old are you?
	• 18-24

	• 25-34 • 35-44 • 45-54 • 55-64 • 65+
5	Choose one or more races/ethnicities that you consider yourself to be
	• White or Caucasian • Black or African American • American Indian/Alaska Native • Asian • Other
6	Are you of Spanish, Hispanic, or Latino origin?
	• Yes • No
7	What is the highest level of education you have completed?
	• Some high school • High school graduate • Some college • Bachelor's degree • Master's degree • Doctoral degree
8	In which country do you currently reside?
	[List of options]
	Attention check
9	We want to know if you are reading this question, please select 'Blue' as your answer
	• Green • Purple • Orange • Blue • Red
	Pre-treatment covariates
	Trust in experts and influencers
10 a	Please indicate how much you agree or disagree with the following statements.
10	I generally trust advice from people with strong knowledge in a topic area
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree

	• Strongly agree
	I often trust health or nutrition advice shared by social media creators
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree • Prefer not to say
11	When sources disagree, I am more likely to trust those with relevant knowledge and training.
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
12	People with specialized knowledge often make issues sound more complicated than they are
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
14	Personal experience can sometimes be more useful than formal qualifications
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
Personal focus on health and nutrition	
15	I pay attention to information about health and nutrition
	• Strongly Disagree • Disagree • Somewhat disagree

	• Neutral • Somewhat agree • Agree • Strongly agree
16	I actively try to improve my health habits (for example, sleep, diet, or exercise)
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
17	I often seek out health or wellness information (online or offline)
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
Health literacy/confidence	
18	I feel comfortable making sense of health-related information.
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
19	I usually have a good sense of what health advice fits my needs.
	• Strongly Disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
Use of social media for health/nutrition	
Where do you get information about health and nutrition (select all that apply)	
	• My doctor • My nutritionist

	• Friends/family • Social media • Other
	Political alignment
	In general, how would you describe your political views?
	• Extremely liberal • Liberal • Slightly liberal • Moderate/middle of the road • Slightly conservative • Conservative • Extremely conservative
	Attention check
20	To show you are paying attention, please select “Strongly Disagree” for this question.
	• Strongly disagree • Disagree • Somewhat disagree • Neutral • Somewhat agree • Agree • Strongly agree
	Treatment and measurement of treatment effect
21 a	Please review this social media creator’s bio [respondent receives one of the posts below]
	 dr.jordankellogg    Dr. Jordan Kellogg  1,498 posts 46.3K followers 920 following Doctor  Harvard & Stanford-trained Gastroenterologist, San Francisco Bay Area  Founder @snackshot.app... more  Jordan Kellogg  1,939 posts 1.1M followers 1,665 following Coach  Health + Wellness Lifestyle  Founder @liveelevatedlife  Coach @joinladder  Unlock Your Potential... more liveelevatedlife.com/summ... and 2
21 b	Imagine you came across the following post while scrolling through social media

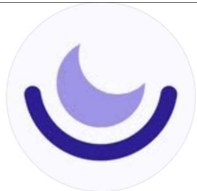
	 dr.jordankellogg ...  ♡ 43.5K 💬 573 🔗 430 🚩 13.6K 🔖 dr.jordankellogg If you want to lower your risk of cancer, avoid eating red meat. Red meat probably causes cancer. Frying or grilling red meat forms carcinogens that increase risk of colorectal, breast, lung, and liver cancer. This means beef but also pork, venison, or lamb. Limiting red meat and swapping in fish, chicken, tofu, or beans is a simple yet effective thing you can do to lower your cancer risk.
	 jordanliftsdaily ...  ♡ 193K 💬 2,466 🔗 4,764 🚩 23.7K 🔖 jordanliftsdaily If you want to lower your risk of cancer, avoid eating red meat. Red meat probably causes cancer. Frying or grilling red meat forms carcinogens that increase risk of colorectal, breast, lung, and liver cancer. This means beef but also pork, venison, or lamb. Limiting red meat and swapping in fish, chicken, tofu, or beans is a simple yet effective thing you can do to lower your cancer risk.
22	What was the main topic of the post you just read: • Red meat and cancer risk • The benefits of grilling vegetables • Fish consumption in children • The importance of eating early • No sure / don't know
23 a	Please answer the following questions about the post you just read.
23	How accurate is the post you just read? • 1 - Not at all accurate • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely accurate

24	How authentic is the post you just read?
	• 1 - Not at all authentic • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely authentic
25	How believable is the post you just read?
	• 1 - Not at all believable • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely believable
	Please briefly describe why you rating this post the level of accuracy, authenticity, and believability that you did. [question only for the pilot]
	[open ended]
26 a	Please review this social media creator's bio
	 stayhydrated406 ... Hydra Hydrated 1 post 2 followers 0 following Hydration made easy Simple water habits for busy days Keep it simple every day
26 b	Imagine you came across the following post while scrolling through social media

	Hydration made easy Simple water habits for busy days Keep it simple every day  stayhydrated406 ...   65  1   stayhydrated406 Drink consistently throughout the day, especially if you are moving. Water plays a key role in keeping your energy up. Without enough, you may feel tired and less able to stay active or pay attention. Easy habits: drink water with meals, carry a water bottle so access is easy, and add high-water foods (like fruits and vegetables) to your diet. Stay hydrated today!	
26 b	Please answer the following questions about the post you just read.	
27	How accurate is the post you just read?	
	• 1 - Not at all accurate • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely accurate	
28	How authentic is the post you just read?	
	• 1 - Not at all authentic • 2 • 3 • 4 - neutral	

	• 5 • 6 • 7 – Extremely authentic		
29	How believable is the post you just read?		
	• 1 - Not at all believable • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely believable		
30 a	Please review this social media creator's bio [respondent receives one of the posts below]		
	<table> <tr> <td>  Dr. Priya Desai ✓ 1,498 46.3K 920 posts followers following Pediatrician Board Certified Pediatrician. Harvard College '07, Stanford School of Medicine '14, and Northwestern Pediatrics Residency '17. www.ohanapediatrics.com </td> <td>  PriyaWellnessDaily ✓ 1,939 1.1M 1,665 posts followers following Fitness Trainer Helping women gain strength + get results that last But also, realistic mom life 💖 @bodylura @vitaluralabs... more join.bodylura.com and 1 more </td> </tr> </table>	 Dr. Priya Desai ✓ 1,498 46.3K 920 posts followers following Pediatrician Board Certified Pediatrician. Harvard College '07, Stanford School of Medicine '14, and Northwestern Pediatrics Residency '17. www.ohanapediatrics.com	 PriyaWellnessDaily ✓ 1,939 1.1M 1,665 posts followers following Fitness Trainer Helping women gain strength + get results that last But also, realistic mom life 💖 @bodylura @vitaluralabs... more join.bodylura.com and 1 more
 Dr. Priya Desai ✓ 1,498 46.3K 920 posts followers following Pediatrician Board Certified Pediatrician. Harvard College '07, Stanford School of Medicine '14, and Northwestern Pediatrics Residency '17. www.ohanapediatrics.com	 PriyaWellnessDaily ✓ 1,939 1.1M 1,665 posts followers following Fitness Trainer Helping women gain strength + get results that last But also, realistic mom life 💖 @bodylura @vitaluralabs... more join.bodylura.com and 1 more		
30 b	Imagine you came across the following post while scrolling through social media [respondent receives one of the posts below]		

	Pediatrician Focus on preventative care Nutrition habits that support growth and well-being Clinic-based care  dr.priyadesai ...  ♥ 12K 💬 277 ↻ 36 🚩 dr.priyadesai Are seed oils bad for kids? Let's break this down. Seed oils contain linoleic acid, an omega-6 fat that is essential for our bodies. We don't make it and need to get it from food. Seed oils do show up in processed foods and snacks. But oil isn't the problem, it's the high nutrient density and lack of whole foods. Swapping out oil doesn't fix that. Focus on what you can add -- fruits and vegetables -- not what you can take away.	Wellness routines Sleep + food habits Simple health swaps New posts daily  priyawellnessdaily ...  ♥ 627K 💬 4,575 ↻ 9,089 🚩 122K priyawellnessdaily Are seed oils bad for kids? Let's break this down. Seed oils contain linoleic acid, an omega-6 fat that is essential for our bodies. We don't make it and need to get it from food. Seed oils do show up in processed foods and snacks. But oil isn't the problem, it's the high nutrient density and lack of whole foods. Swapping out oil doesn't fix that. Focus on what you can add -- fruits and vegetables -- not what you can take away.
31	What was the main topic of the post you just read:	
	• Seed oils and kids' diet • The nutritional value of whole wheat bread • School lunch costs • Meal schedules for toddlers • No sure / don't know	
32 a	Please answer the following questions about the post you just read.	
32	How accurate is the post you just read?	
	• 1 - Not at all accurate • 2 • 3 • 4 - neutral	

	• 5 • 6 • 7 – Extremely accurate
33	How authentic is the post you just read?
	• 1 - Not at all authentic • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely authentic
34	How believable is the post you just read?
	• 1 - Not at all believable • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely believable
	Why did you rate the post the level of accurate, authentic, and believable/trustworthy that you did? [question only for the pilot]
	[open ended]
35 a	Please review this social media creator's bio
	 sleepwell821 ... KK 2 posts 1 follower 0 following Focused on better sleep Sleep habits made simple Always thinking about the zzzz's
35 b	Imagine you came across the following post while scrolling through social media

	Focused on better sleep Sleep habits made simple Always thinking about the zzzz's  sleepwell821 ...  Set limits on news & social media before bed  Keep screens out of the bedroom at night  Create a calming wind-down routine  Jot down lingering thoughts before bed to clear your mind  Take slow breaths to help settle your body  Make time for hobbies that help you unwind, like drawing, reading, stretching, or calling a friend  Go to bed & wake up at the same time each day  65  1    sleepwell821 For a good night's sleep, learn how to manage stress: set limits on news & social media before bed, keep screens out of the bedroom at night, create a calming wind-down routine, and take slow breaths to settle your body. Make time for hobbies that help you unwind, and go to bed and wake up at the same time each day. There's a lot you can do make sure you get a good night's sleep.	
35 b	Please answer the following questions about the post you just read.	
36	How accurate is the post you just read?	
	• 1 - Not at all accurate • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely accurate	
37	How authentic is the post you just read?	
	• 1 - Not at all authentic • 2	

	• 3 • 4 - neutral • 5 • 6 • 7 – Extremely authentic
38	How believable is the post you just read?
	• 1 - Not at all believable • 2 • 3 • 4 - neutral • 5 • 6 • 7 – Extremely believable